SOKOINE UNIVERSITY OF AGRICULTURE

SCHOOL OF ENGINEERING AND TECHNOLOGY (SoET)

DEPARTMENT OF AGRICULTURAL ENGINEERING (DAE)

AE 219

ASSIGNMENT 6

Note:

[1] Please listen to videos https://www.youtube.com/playlist?list=PLmKnXyyknMkPhutqZ6uebPazrUCobJ8Ep provided by my Lab YEESI Lab and instructed by Dr Masamaki

[2] Please use Google Colab: https://colab.research.google.com/

[3] Please read and listen to the videos at https://towardsdatascience.com/best-time-saving-python-tricks-in-jupyter-notebook-fca877132507

[4] Please read how to input data into a program: https://www.askpython.com/python/examples/python-user-input

Questions (All programs should run on the command line. Read Instruction 4):

1. Suppose you are managing a large farm with different crops, and you want to analyze the distribution of crops across various fields. Each field is represented by a unique identifier, and the crops planted in each field are stored in a list. Additionally, each crop is represented as a dictionary with attributes such as crop name, yield (in tons), and price per ton.
   Write a Python program that takes the following inputs:
   a. A list of fields, where each field is represented by a dictionary with attributes: 'field_id' (a unique identifier for the field), 'location' (a string representing the field's location), and 'crops' (a list of dictionaries representing the crops planted in that field).
   b. A target crop name, for which you want to calculate the total yield and revenue across all fields.
   Your program should perform the following tasks:
   i. Iterate through the list of fields and find the fields where the target crop is planted.
   ii. For each field with the target crop, calculate the total yield (sum of yields from all crops of that type) and revenue (yield multiplied by price per ton).

iii. Finally, display the total yield and revenue for the target crop across all fields.

Provide your Python code to solve this problem and demonstrate its functionality using a sample dataset.

Note: Feel free to create a realistic sample dataset with multiple fields and crops to showcase the functionality of your solution.

2. Consider a dataset containing information about different crops grown in a region. Each crop is represented by a dictionary with the following keys: 'crop_name', 'yield', 'average_temperature', and 'rainfall'. You are given a list named 'crop_data' that contains several dictionaries, each representing a different crop.
Your task is to write a Python function named 'top_yield_crops' that takes in the 'crop_data' list as a parameter and returns a list of the top three crops with the highest yield. The yield is given in kilograms per hectare. However, there are a few conditions to consider while selecting the top crops:
    a. The average temperature during the crop's growth period should be between 20 to 30 degrees Celsius (inclusive).
    b. The rainfall during the crop's growth period should be between 500 to 1000 millimetres (inclusive).

   Additionally, you need to account for the possibility of missing or invalid data. If a crop's dictionary is missing any of the required keys or if the values are not numeric, it should be skipped while determining the top-yield crops.
   Note: You can assume that the 'crop_data' list is not empty and contains at least three crops with valid data.
3. Consider a dataset containing information about crop yields in different regions. The dataset is represented as a list of dictionaries, where each dictionary represents a region and contains the following keys: 'region', 'crop', 'year', and 'yield'.
   Here's an example dataset:

```
crop_yield_data = [
    {'region': 'Region A', 'crop': 'Corn', 'year': 2019, 'yield': 100},
    {'region': 'Region A', 'crop': 'Wheat', 'year': 2019, 'yield': 200},
    {'region': 'Region A', 'crop': 'Corn', 'year': 2020, 'yield': 150},
    {'region': 'Region A', 'crop': 'Wheat', 'year': 2020, 'yield': 250},
    {'region': 'Region B', 'crop': 'Corn', 'year': 2019, 'yield': 120},
    {'region': 'Region B', 'crop': 'Wheat', 'year': 2019, 'yield': 180},
    {'region': 'Region B', 'crop': 'Corn', 'year': 2020, 'yield': 130},
    {'region': 'Region B', 'crop': 'Wheat', 'year': 2020, 'yield': 220},
]
```

   Write a Python function called get_average_yield_by_crop that takes in the dataset and a crop name as input and returns the average yield for that crop across all regions and years. The function should consider only the records that match the given crop name.

For example, get_average_yield_by_crop(crop_yield_data, 'Corn') should return 116.66666666666667 (rounded to 2 decimal places), and get_average_yield_by_crop(crop_yield_data, 'Wheat') should return 212.5 (rounded to 2 decimal places).

Note: You can assume that the dataset will always contain at least one record for each combination of region, crop, and year.

Submit your code results by Tuesday midnight (4-June-2023) to the CR as the softcopy and CR will forward the bundle of all codes to me on the next day. DO NOT SEND YOUR WORK to the Instructor's email. For any enquiries, please send an email.

As usual, Every one hour late equates to 5 marks deduction.